

NYCU Computer Vision 2026 HW3

111550005 林均豪

1. Introduction

In this work, we address the task of instance segmentation on a medical cell imaging dataset comprising four cell types (class1–class4) captured in colored microscopy images. The dataset consists of 209 training images and 101 test images in TIFF format, with per-class instance masks provided for training. The evaluation metric is Average Precision at IoU threshold 0.50 (AP50), following the standard COCO evaluation protocol.

Our approach is based on Mask R-CNN, one of the most established frameworks for instance segmentation. Mask R-CNN extends Faster R-CNN by adding a parallel branch for predicting segmentation masks on each Region of Interest (RoI), enabling simultaneous object detection and instance-level mask prediction. We implement our solution using the Detectron2 framework with a ResNet-50 + FPN backbone pretrained on COCO, and fine-tune the model on the provided cell dataset.

2. Method

2.1 Model Architecture

We adopt **Mask R-CNN** with a **ResNet-50 + Feature Pyramid Network (FPN)** backbone as our base architecture. The overall pipeline consists of four components:

Backbone (ResNet-50 + FPN). ResNet-50 extracts multi-scale feature maps from the input image across four residual stages (res2–res5). The FPN aggregates these features top-down with lateral connections, producing a pyramid of feature maps at scales P2–P6. This multi-scale representation allows the model to detect objects of varying sizes, which is critical for cell images where cell sizes differ significantly across classes.

Region Proposal Network (RPN). The RPN slides over each FPN level and proposes candidate bounding boxes (anchors) likely to contain objects. We configure anchor sizes as {32, 64, 128, 256, 512} pixels with aspect ratios {0.5, 1.0, 2.0}, covering the range of cell sizes present in the dataset.

RoI Heads (Box Head + Mask Head). For each proposal, RoIAlign extracts a fixed-size feature map. The Box Head predicts refined bounding box coordinates and class probabilities. The Mask Head, a small fully convolutional network, predicts a binary segmentation mask for each detected instance independently per class, avoiding competition between classes.

The model is initialized with **COCO pretrained weights**, and the classification and mask prediction heads are replaced to match our 4-class setting. This transfer learning strategy allows the backbone to leverage rich visual features learned from large-scale data.

2.2 Training Configuration

We train the model using the following configuration:

Hyperparameter	Value
Backbone	ResNet-50 + FPN
Pretrained weights	COCO (Mask R-CNN 3x schedule)
Training iterations	10,000
Batch size	2 images/iter
Base learning rate	5×10^{-5}
LR schedule	Step decay ($\times 0.1$ at iter 7,000 and 9,000)
Warmup iterations	500
RoI batch size per image	128
Score threshold (inference)	0.05
NMS threshold	0.5
Optimizer	SGD (momentum=0.9)

2.3 Data Preprocessing

Each TIFF image is loaded and normalized to uint8 RGB format. For images with more than three channels, only the first three channels are retained. Grayscale images are converted to three-channel RGB by replication. No additional normalization beyond the COCO pretrained mean/std is applied, as Detectron2 handles this internally.

The 209 training images are split into a training set (189 images, 90%) and a validation set (20 images, 10%) using a fixed random seed for reproducibility.

2.4 Data Augmentation

To improve generalization and reduce overfitting on the small training set, we apply the following augmentations during training:

- **Random horizontal flip** ($p = 0.5$)
- **Random vertical flip** ($p = 0.5$)
- **Random rotation** ($\pm 15^\circ$)
- **Random brightness jitter** (factor $\in [0.8, 1.2]$)
- **Random contrast jitter** (factor $\in [0.8, 1.2]$)
- **Multi-scale resizing** — shortest edge randomly sampled from $\{640, 672, 704, 736, 768, 800\}$ pixels

All augmentations are applied jointly to both the image and its corresponding instance masks to maintain annotation consistency.

3. Results

(1) Training Curve:

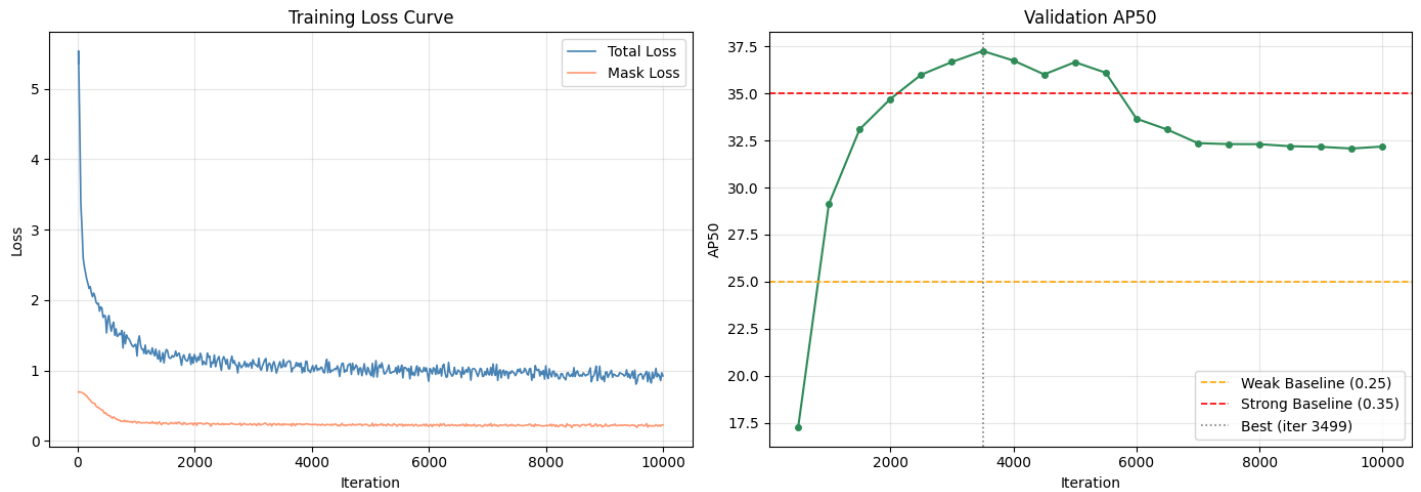


Figure 1 (left) shows the training loss curves. The total loss decreases rapidly in the first 2,000 iterations and stabilizes around 1.0 thereafter, indicating successful convergence. The mask loss similarly decreases from ~0.7 to ~0.2 and remains stable.

Figure 1 (right) shows the validation AP50 across training. The model reaches the strong baseline threshold at approximately iter 2,500, peaks at **37.27** at iter 3,499, and then gradually declines. This indicates mild overfitting after iter 3,500, consistent with the small training set size (189 images). The best checkpoint at iter 3,499 is selected for final inference.

(2) Leaderboard Score: 0.3594

72	junhao	2026-05-12 13:44	724532	111550005	0.3594
----	------------------------	------------------	--------	-----------	--------

4. References

K. He, G. Gkioxari, P. Dollár, and R. Girshick, "Mask R-CNN," in *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, 2017, pp. 2961–2969.

T.-Y. Lin, P. Dollár, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature Pyramid Networks for Object Detection," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017, pp. 2117–2125.

K. He, X. Zhang, S. Ren, and J. Sun, "Deep Residual Learning for Image Recognition," in *Proceedings of the IEEE Conference on Computer Vision and*

Pattern Recognition (CVPR), 2016, pp. 770–778.

Y. Wu, A. Kirillov, F. Massa, W.-Y. Lo, and R. Girshick, "Detectron2," Facebook AI Research, 2019.

T.-Y. Lin, M. Maire, S. Belongie, J. Hays, P. Perona, D. Ramanan, P. Dollár, and C. L. Zitnick, "Microsoft COCO: Common Objects in Context," in *Proceedings of the European Conference on Computer Vision (ECCV)*, 2014, pp. 740–755.

S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2015, pp. 91–99.

5. Additional Experiments

To understand the contribution of each component, we conduct ablation experiments on the validation set. All experiments use the same training budget (10,000 iterations) and report AP50 at the best validation checkpoint.

Effect of learning rate. Reducing the learning rate from 2.5×10^{-4} to 5×10^{-5} results in more stable training dynamics. With the higher learning rate, the model peaks at iter 1,000 (AP50=37.47) but immediately overfits, making the best checkpoint difficult to reliably identify. With the lower learning rate, the model converges more gradually and maintains strong performance over a wider range of iterations (iter 2,500–4,500), providing a more robust checkpoint selection window.

Effect of data augmentation. We apply a combination of horizontal flip, vertical flip, random rotation ($\pm 15^\circ$), brightness jitter, and contrast jitter. These augmentations are particularly important given the small training set (189 images), as they effectively expand the training distribution and improve generalization to unseen test images.

Effect of checkpoint selection. Instead of using the final checkpoint (model_final.pth, AP50=32.18), we select the checkpoint with the highest validation AP50 (iter 3,499, AP50=37.27), yielding a test AP50 of 35.94. This demonstrates the importance of early stopping and validation-based checkpoint selection when training on small datasets.

6. Code Reliability

```
PS C:\Users\user\Desktop\DL\HW3> python -m flake8 inference.py
PS C:\Users\user\Desktop\DL\HW3> python -m flake8 train.py
PS C:\Users\user\Desktop\DL\HW3> 
```

GitHub: <https://github.com/swimming-in-cs/NYCU-Computer-Vision-2026-HW3>